UNIT 8 · TOPIC 3

# Common Tools and Press Operations

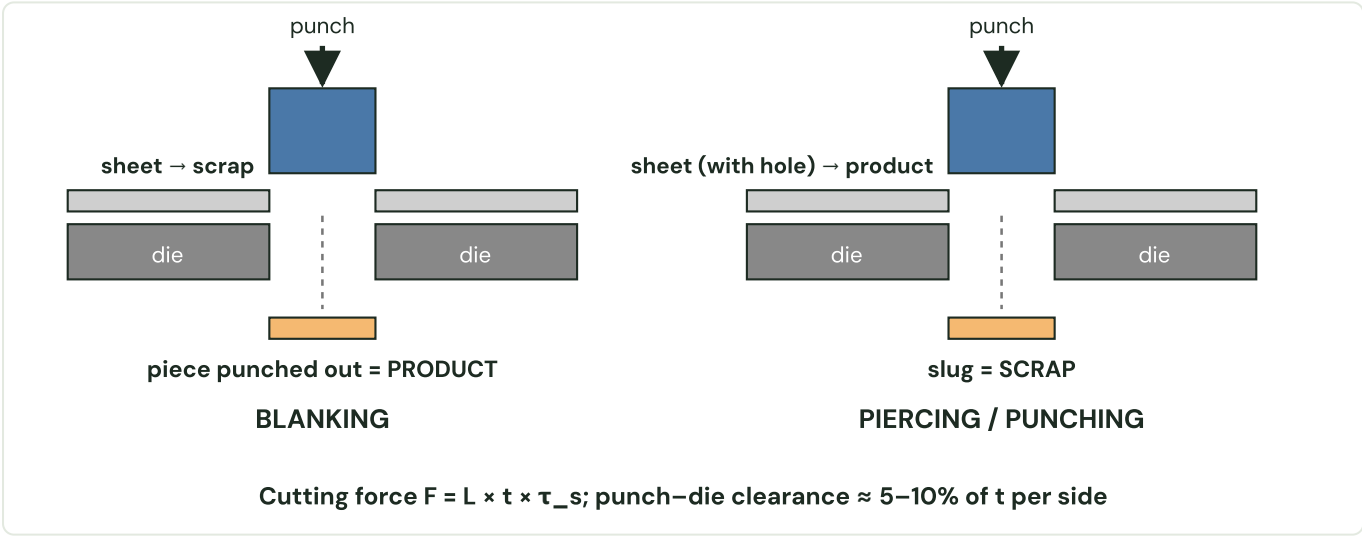
*Syllabus topic: Common tools and press operations*

## Fitting (bench) tools

| Tool | Use |
|---|---|
| Bench vice | Holds the work |
| Files (flat/round/triangular), hacksaw, cold chisels, hammers | Shaping and cutting |
| Punches (centre/pin), scriber, try-square, calipers, rule, marking media | Marking and measuring |
| TAPS | Cut INTERNAL threads |
| DIES | Cut EXTERNAL threads |
| REAMERS | Finish/size an existing hole accurately |
| Snips, stakes, mallets, folding bars | Sheet-metal work |

## Press (sheet-metal) operations

A PRESS drives a punch into a die to cut or form sheet metal.




*Blanking vs piercing (redrawn — the source figure showed six identical punch sketches)*

| Operation | What happens |
|---|---|
| SHEARING | Straight-line cut separating the sheet into pieces |
| BLANKING | The piece punched OUT is the PRODUCT; the sheet with the hole is scrap |
| PIERCING / PUNCHING | The HOLE is wanted; the removed slug is scrap |
| Notching, trimming, lancing | Other cutting operations |
| BENDING | Angular deformation |
| DRAWING (deep drawing) | Makes a cup/shell from a flat blank |
| COINING, EMBOSSING | Impress shapes |
| Stretch forming | Forming over a die while stretched |

$$\text{Cutting force } F = L \times t \times \tau_s$$

L = length of cut (perimeter for blanking); t = sheet thickness; $\tau_s$ = shear strength of the material.

- **Punch–die CLEARANCE ≈ 5–10% of thickness per side**: too little → double break; too much → burr.
- A shear angle on the punch reduces the peak force.
- **Dies:** simple; COMPOUND (blank + pierce in one stroke); PROGRESSIVE (successive stations).

## Practice MCQs

**Q37. A tool used to cut internal threads is a:**

(a) Die (b) Tap
(c) Reamer (d) Scriber

**Q38. External threads on a rod are cut with a:**

(a) Tap (b) Die
(c) Reamer (d) Broach

**Q39. A reamer is used to:**

(a) Drill a new hole (b) Finish/size an existing hole accurately
(c) Cut threads (d) Mark layout

**Q40. In BLANKING, the useful product is the:**

(a) Sheet with the hole (b) Piece punched out
(c) Scrap slug (d) Chip

**Q41. In PIERCING/PUNCHING, the removed slug is:**

(a) The product (b) Scrap
(c) Reused as tool (d) The die

**Q42. Shearing produces a:**

(a) Cup (b) Straight-line cut separating the sheet
(c) Threaded hole (d) Weld

**Q43. Making a cup or shell from a flat blank is:**

(a) Blanking (b) Deep drawing
(c) Piercing (d) Coining

**Q44. Cutting force in blanking is F =**

(a) L × t × τ_s (b) L/t

(c) t/τ_s (d) L + t + τ_s

**Q45. Punch–die clearance in cutting is typically:**

(a) 0% (b) 5–10% of thickness per side

(c) 50% (d) Equal to thickness

**Q46. A compound die performs:**

(a) One operation (b) Two operations (e.g., blank + pierce) in one stroke

(c) Only bending (d) No cutting

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**Answers**

Q37 — (b) Tap

Q38 — (b) Die

Q39 — (b) Finish/size an existing hole accurately

Q40 — (b) Piece punched out — *Blanking → the punched-out piece is the product.*

Q41 — (b) Scrap — *Piercing → the hole is wanted; the slug is scrap — the classic pair.*

Q42 — (b) Straight-line cut separating the sheet

Q43 — (b) Deep drawing

Q44 — (a) L × t × τ_s — *F = L·t·τ_s.*

Q45 — (b) 5–10% of thickness per side — *Clearance 5–10% of thickness per side controls edge quality.*

Q46 — (b) Two operations (e.g., blank + pierce) in one stroke

**Sources:** Hajra Choudhury — Elements of Workshop Technology (Vols I & II); Kalpakjian — Manufacturing Engineering & Technology; standard production-engineering texts; NPTEL manufacturing lectures.